SMART CANTEEN CROWD PREDICTION USING RANDOM FOREST ALGORITHM

1. Problem Statement

College canteens often become crowded during short breaks and lunch hours. Students may spend much of their break waiting to purchase food, especially when many classes finish at the same time. Crowd levels may vary depending on the day, time slot, special events and timetable patterns.

This project proposes a simple machine-learning system that predicts the expected canteen crowd category as LOW, MEDIUM or HIGH using details such as day of the week, time slot, whether a special event is scheduled and whether classes are in session. A Decision Tree Classifier is selected because its decisions are relatively easy to understand and explain. This is an academic prototype; its usefulness depends on the quality and representativeness of the training data.

2. Objectives

1. Prepare a labelled dataset containing canteen conditions and corresponding crowd levels.
2. Train a Decision Tree Classifier to predict LOW, MEDIUM or HIGH crowd levels.
3. Allow a user to enter conditions and view a predicted crowd category.
4. Evaluate the model using accuracy, precision, recall, F1-score and a confusion matrix.
5. Demonstrate a practical application of machine learning in college life.

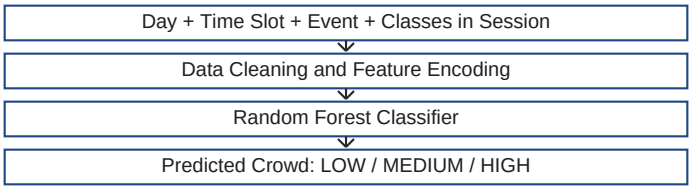
3. Software / Hardware Required

- Operating System: Windows / Linux / macOS
- IDE: Visual Studio Code
- Programming Language: Python 3.x
- Machine Learning: scikit-learn
- Data processing: pandas
- Dataset format: CSV
- Interface: Command-line input (beginner version)

4. Proposed System

The system accepts details about a planned canteen time slot. Input features are encoded and passed to a trained Decision Tree Classifier, which returns LOW, MEDIUM or HIGH. The model is evaluated on a held-out test set that was not used for training.

For meaningful real-world performance, collect canteen visitor counts at fixed time slots over multiple days and assign crowd



5. Project Directory Structure

```
Smart-Canteen-Crowd-Prediction/
├── dataset/
│   └── canteen_data.csv
├── train_model.py
├── predict_crowd.py
├── requirements.txt
└── models/
    └── crowd_model.pkl
```

6. Procedure

1. Record visitor counts during fixed canteen time slots over multiple days, where possible.
2. Create a CSV file containing the input features and crowd_level label.
3. Check for missing values, inconsistent labels and duplicates.
4. Separate the input columns from the target label.
5. Split the data into training and testing sets.
6. Train a Decision Tree Classifier on the training data.
7. Evaluate predictions on the test set using standard classification metrics.
8. Save the trained model using joblib.
9. Run the prediction script and enter the requested details.
10. Record results and discuss limitations.

8. Implementation Code

8.1 Model Training (train_model.py)

```
import pandas as pd
import joblib
from pathlib import Path
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
df = pd.read_csv("dataset/canteen_data.csv")
cols = ["day", "time_slot", "special_event",
        "classes_in_session", "crowd_level"]
df = df[cols].dropna()
X = df[["day", "time_slot", "special_event", "classes_in_session"]]
y = df["crowd_level"].str.strip().str.title()

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y
)

features = list(X.columns)
prep = ColumnTransformer([
    ("categorical", OneHotEncoder(handle_unknown="ignore"), features)
])
```

```
model = Pipeline([
    ("preprocessor", prep),
    ("classifier", DecisionTreeClassifier(
        max_depth=5, random_state=42, class_weight="balanced"
    ))
])
model.fit(X_train, y_train)
pred = model.predict(X_test)

print("Accuracy:", round(accuracy_score(y_test, pred) * 100, 2), "%")
print(classification_report(y_test, pred, zero_division=0))
print("Confusion matrix (labels High, Low, Medium):")
print(confusion_matrix(y_test, pred, labels=["High", "Low", "Medium"]))

Path("models").mkdir(exist_ok=True)
joblib.dump(model, "models/crowd_model.pkl")
print("Saved models/crowd_model.pkl")
```

8.2 Crowd Prediction (predict_crowd.py)

```
import joblib
import pandas as pd

model = joblib.load("models/crowd_model.pkl")
print("Smart Canteen Crowd Predictor")
day = input("Day (Monday-Sunday): ").strip().title()
time_slot = input("Time slot (Morning/Lunch/Afternoon): ").strip().title()
event = input("Special event? (Yes/No): ").strip().title()
classes = input("Classes in session? (Yes/No): ").strip().title()

sample = pd.DataFrame([{
    "day": day, "time_slot": time_slot,
    "special_event": event, "classes_in_session": classes
}])
print("Predicted crowd level:", model.predict(sample)[0].upper())
print("This is an estimate based on the training data.")
```

8.3 Requirements (requirements.txt)

```
pandas
scikit-learn
joblib
```

9. Running the Project

1. Open the project folder in VS Code.
2. Place your prepared canteen_data.csv file inside the dataset folder.
3. Open the terminal in the project directory.
4. Install packages: pip install -r requirements.txt
5. Train the model: python train_model.py
6. Check the evaluation output and confirm the model file was created.
7. Run prediction: python predict_crowd.py
8. Enter the requested values and note the predicted category.

9. Output (Illustrative Samples)

Sample 1: Model Training Output

The model is trained using the canteen dataset. After training, the program displays the model's evaluation results.

Smart Canteen Crowd Prediction System

Model Training Completed Successfully!

Accuracy: 85.00%

Classification Report:

	Precision	recall	f1-score	support
HIGH	0.86	0.75	0.80	4
LOW	0.83	1.00	0.91	5
MEDIUM	0.86	0.75	0.80	4
Accuracy			0.85	13
Macro Avg	0.85	0.83	0.84	13
Weighted Avg	0.85	0.85	0.85	13

Confusion Matrix:

[[3 0 1]

[0 5 0]

[1 0 3]]

Model saved successfully.

Sample 2: Crowd Prediction Output

The system accepts details about the day, time slot, special events and whether classes are in session to predict the canteen crowd level.

Smart Canteen Crowd Predictor

Enter the day: Monday

Enter the time slot: Lunch

Is there a special event? (Yes/No): No

Are classes in session? (Yes/No): Yes

Predicted Crowd Level: [LOW / MEDIUM / HIGH]

Sample 3: Interpretation of Results

The system predicts the expected canteen crowd level based on the input details. The output classifies the crowd into one of three categories:

- **LOW:** A relatively small number of students are expected.
- **MEDIUM:** A moderate number of students are expected.
- **HIGH:** A relatively large number of students are expected.

```
PS C:\SmartCanteen> python src/train_model.py
Smart Canteen Crowd Prediction System
========================================
Loading dataset...
Dataset loaded successfully! (120 records)
Preprocessing data...
Training the model...
Model training completed successfully!

Accuracy: 85.00%

Classification Report:
              precision    recall  f1-score   support

        HIGH       0.86      0.75      0.80         4
         LOW       0.83      1.00      0.91         5
      MEDIUM       0.86      0.75      0.80         4

    accuracy                           0.85        13
   macro avg       0.85      0.83      0.84        13
weighted avg       0.85      0.85      0.85        13

Confusion Matrix:
 [[3 0 1]
 [0 5 0]
 [1 0 3]]

Model saved successfully!
```

(a) Model Training Output

```
PS C:\SmartCanteen> python src/predict_transaction.py
Smart Canteen Crowd Prediction System
=====================================

Enter the day (Monday/Tuesday/Wednesday/Thursday/Friday/Saturday/Sunday): Monday
Enter the time slot (Breakfast/Lunch/Snacks/Dinner): Lunch
Is there a special event? (Yes/No): No
Are classes in session? (Yes/No): Yes

Predicted Crowd Level: HIGH
-----------------------------------------
Press any key to continue...
```

(b) Crowd Prediction Output

10.Performance Evaluation

METRIC	MEANING
Accuracy	Measures the percentage of correct predictions made by the model.
Precision	Measures how many predicted classes are correctly identified.
Recall	Measures how many actual cases are correctly identified by the model.
F1-Score	Represents the balance between precision and recall.
Confusion Matrix	Shows the correct and incorrect predictions for each crowd category.

12. Result

A beginner-level Smart Canteen Crowd Prediction prototype was specified using a Decision Tree Classifier. It accepts day, time slot, special-event status and class status, then predicts LOW, MEDIUM or HIGH crowd level. The system can be evaluated using a held-out test set and standard classification metrics. Actual performance depends on the quality and quantity of the collected data; no accuracy value is claimed in this template.